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Inventor(s)	Boyapalle; Anantha K. et al.

Method and apparatus for edge storage for enterprise applications

Abstract

An information handling system comprising a processor, a memory device, and a power management unit (PMU). The processor to execute computer readable program code of an interim edge storage management system, the processor operatively coupled to a client device, an edge storage device, and a plurality of remote network file hosting systems managed by the interim edge storage management system, wherein the interim edge storage management system includes a storage orchestrator to manage data storage of the client device at and access to the edge storage device and the plurality of remote network file hosting systems based on recency of usage of files to be stored.

Inventors: Boyapalle; Anantha K. (Cedar Park, TX), Quinn; Liam B. (Austin, TX), Schwebke; Curtis J. (Los Gatos, CA)

Applicant: Dell Products, LP (Round Rock, TX)

Family ID: 89509781

Assignee: DELL PRODUCTS LP (Round Rock, TX)

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Primary Examiner: Doan; Tan

Attorney, Agent or Firm: Prol Intellectual Property Law, PLLC

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to data storage. The present disclosure more specifically relates to optimizing data storage of client device files on one or more file hosting systems.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to clients is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing clients to take advantage of the value of the information. Because technology and information

handling needs and requirements vary between different clients or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific client or specific use, such as e-commerce, financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems. The information handling system may include telecommunication, network communication, and video communication capabilities. Network communications may allow a user to access a file storage service such as a file hosting system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:
- (2) FIG. 1 is a block diagram of an information handling system according to an embodiment of the present disclosure;
- (3) FIG. 2 is a block diagram of a network environment offering several communication protocol options and mobile information handling systems according to an embodiment of the present disclosure;
- (4) FIG. 3 is a block diagram of a network edge server executing an interim edge storage management system in an embodiment of the present disclosure;
- (5) FIG. 4 is a block process diagram describing a method of operating an interim edge storage management system according to an embodiment of the present disclosure;
- (6) FIG. 5 is a flow diagram of a method of an information handling system operatively coupled to an edge device executing an interim edge storage management system according to an embodiment of the present disclosure;
- (7) FIG. 6 is a flow diagram of a method of executing an interim edge storage management system according to an embodiment of the present disclosure; and
- (8) FIG. 7 is a flow diagram of a method of executing one or more network file hosting systems with an interim edge storage management system according to an embodiment of the present disclosure.
- (9) The use of the same reference symbols in different drawings may indicate similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

- (10) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings, and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.
- (11) Information handling systems allow a system to communicate data with other systems that are remote from the information handling system. Further, files may be shared among a plurality of client information handling systems, in an enterprise for example, in some embodiments. Part of this communication includes the sharing of files to an online network file hosting system for the

information handling system or other authorized information handling system to access those files from multiple locations. These network file hosting systems, however, have data storage limits. For example, customers of OneDrive® by Microsoft® may be limited to 1 terabyte (1 TB) per user. These customers may be able to increase the data storage (e.g., 5 TB) with an increased subscription cost. Any additional storage capacities may be limited to the users by administrators of the network file hosting system and may be granted on a case-by-case basis. These extra steps may be cumbersome to the user or an administrator and may increase the costs of operation for the user. Still further, latency issues may arise when a user is attempting to access the files stored on the network file hosting system that may be located at a distant server.

(12) The present specification describes an edge device with storage for every user within, for example, an enterprise. This edge device may be an interim edge storage management system that is operatively coupled to a client device (e.g., information handling system) and a network file hosting system. This interim edge storage management system may include a storage orchestrator that manages data storage of the client device at the network file hosting system. In an embodiment, the storage orchestrator may be operatively coupled to a cloud computing management console that provides active directory synchronization files from a local active directory hosted on a virtual machine that describes how the storage orchestrator manages data storage for the client device between local storage, edge storage, and remote network file hosting systems.

(13) In an embodiment, the active directory synchronization files may include data descriptive of which files from the client devices are stored among local storage, edge storage, or remote network file hosting systems. For example, the active directory synchronization files may be received by the storage orchestrator and the storage orchestrator may store certain files in a virtual network-attached edge storage device for low-latency access to these files by the client device or a plurality of client devices. In an embodiment, the active directory synchronization files may be received by the storage orchestrator and the storage orchestrator may save one or more files in one or more network file hosting systems. This may include the storage orchestrator saving these files among storage locations based on the most recently used (MRU) files. For example, those files most recently accessed, updated, saved, or otherwise manipulated by a client device may be stored on the network file hosting systems.

(14) In an embodiment, the storage orchestrator may save these MRU files to a single network file hosting system. In an embodiment, the storage orchestrator may save these files to a plurality of network file hosting systems using, for example, a round robin scheduling protocol that saves a first MRU file to a first network file hosting system, a second MRU file to a second network file hosting system, a third MRU file to another network file hosting system and so on. In an embodiment, this process may allow the storage orchestrator to store the MRU files until a lowest level of data storage capacity is met on any network file hosting system. When the lowest level of data storage capacity is met on any network file hosting system, the storage orchestrator may engage in a first-in, first-out process by deleting a file from a network file hosting system before saving a new MRU to that network file hosting system.

(15) In an embodiment, the storage orchestrator may execute, via a processing device, one or more application program interfaces (APIs) that allow the storage orchestrator to interface with the network file hosting systems described herein. In an embodiment, the storage orchestrator may be a micro-service of one of the network file hosting systems associated with the interim edge storage management system.

(16) FIG. 1 illustrates an information handling system **100** similar to information handling systems according to several aspects of the present disclosure. In the embodiments described herein, an information handling system **100** includes any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or use any form of information, intelligence, or data for

business, scientific, control, entertainment, or other purposes. For example, an information handling system **100** can be a personal computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a consumer electronic device, a network server or storage device, a network router, switch, or bridge, wireless router, or other network communication device, a network connected device (cellular telephone, tablet device, etc.), IoT computing device, wearable computing device, a set-top box (STB), a mobile information handling system, a palmtop computer, a laptop computer, a desktop computer, a convertible laptop, a tablet, a smartphone, a communications device, an access point (AP), a base station transceiver, a wireless telephone, a control system, a camera, a scanner, a printer, a personal trusted device, a web appliance, or any other suitable machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine, and can vary in size, shape, performance, price, and functionality.

(17) In a networked deployment, the information handling system **100** may operate in the capacity of a server or as a client computer in a server-client network environment, or as a peer computer system in a peer-to-peer (or distributed) network environment. In a particular embodiment, the computer system **100** can be implemented using electronic devices that provide voice, video, or data communication. For example, an information handling system **100** may be any mobile or other computing device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. In an embodiment, the information handling system **100** may be operatively coupled to a server or other network device as well as with an edge device used to orchestrate the sharing of files locally, at the edge, or in the cloud. Further, while a single information handling system **100** is illustrated, the term “system” shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or multiple sets, of instructions to perform one or more computer functions.

(18) The information handling system **100** may include memory (volatile (e.g., random-access memory, etc.), nonvolatile (read-only memory, flash memory etc.) or any combination thereof), one or more processing resources, such as a central processing unit (CPU), a graphics processing unit (GPU) **152**, processing, hardware, controller, or any combination thereof. Additional components of the information handling system **100** can include one or more storage devices, one or more communications ports for communicating with external devices, as well as, various input and output (I/O) devices **140**, such as a keyboard **144**, a mouse **150**, a video display device **142**, a stylus **146**, a trackpad **148**, or any combination thereof. The information handling system **100** can also include one or more buses **116** operable to transmit data communications between the various hardware components described herein. Portions of an information handling system **100** may themselves be considered information handling systems and some or all of which may be wireless.

(19) Information handling system **100** can include devices or modules that embody one or more of the devices or execute instructions for the one or more systems and modules described above, and operates to perform one or more of the methods described above. The information handling system **100** may execute code instructions **110** via processing resources that may operate on servers or systems, remote data centers, or on-box in individual client information handling systems according to various embodiments herein. In some embodiments, it is understood any or all portions of code instructions **110** may operate on a plurality of information handling systems **100**.

(20) The information handling system **100** may include processing resources such as a processor **102** such as a central processing unit (CPU), accelerated processing unit (APU), a neural processing unit (NPU), a vision processing unit (VPU), an embedded controller (EC), a digital signal processor (DSP), a GPU **152**, a microcontroller, or any other type of processing device that executes code instructions to perform the processes described herein. Any of the processing resources may operate to execute code that is either firmware or software code. Moreover, the information handling system **100** can include memory such as main memory **104**, static memory **106**, computer readable medium **108** storing instructions **110** of, in an example embodiment, a

video conferencing application, an image capturing application, or other computer executable program code, and drive unit **118** (volatile (e.g., random-access memory, etc.), nonvolatile (read-only memory, flash memory etc.) or any combination thereof).

(21) As shown, the information handling system **100** may further include a video display device **142**. The video display device **142**, in an embodiment, may function as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, or a solid-state display. Although FIG. **1** shows a single video display device **142**, the present specification contemplates that multiple video display devices **142** may be used with the information handling system to facilitate an extended desktop scenario, for example. Additionally, the information handling system **100** may include one or more input/output devices **140** including an alpha numeric input device such as a keyboard **144** and/or a cursor control device, such as a mouse **150**, touchpad/trackpad **148**, a stylus **146**, or a gesture or touch screen input device associated with the video display device **142** that allow a user to interact with the images, windows, and applications presented to the user. In an embodiment, the video display device **142** may provide output to a user that includes, for example, one or more windows describing one or more instances of applications being executed by the processor **102** of the information handling system. In this example embodiment, a window may be presented to the user that provides a graphical user interface (GUI) representing the execution of that application.

(22) The network interface device of the information handling system **100** shown as wireless interface adapter **126** can provide connectivity among devices such as with Bluetooth or to a network **134**, e.g., a wide area network (WAN), a local area network (LAN), wireless local area network (WLAN), a wireless personal area network (WPAN), a wireless wide area network (WWAN), or other network. In an embodiment, the WAN, WWAN, LAN, and WLAN may each include an access point **136** or base station **138** used to operatively couple the information handling system **100** to a network **134** and, in an embodiment, to an interim edge storage management system **154** described herein. In a specific embodiment, the network **134** may include macro-cellular connections via one or more base stations **138** or a wireless access point **136** (e.g., Wi-Fi or WiGig), or such as through licensed or unlicensed WWAN small cell base stations **138**.

Connectivity may be via wired or wireless connection. For example, wireless network access points **136** or base stations **138** may be operatively connected to the information handling system **100**. Wireless interface adapter **126** may include one or more radio frequency (RF) subsystems (e.g., radio **128**) with transmitter/receiver circuitry, modem circuitry, one or more antenna front end circuits **130**, one or more wireless controller circuits, amplifiers, antennas **132** and other circuitry of the radio **128** such as one or more antenna ports used for wireless communications via multiple radio access technologies (RATs). The radio **128** may communicate with one or more wireless technology protocols. In an embodiment, the radio **128** may contain individual subscriber identity module (SIM) profiles for each technology service provider and their available protocols for any operating subscriber-based radio access technologies such as cellular LTE communications.

(23) In an example embodiment, the wireless interface adapter **126**, radio **128**, and antenna **132** may provide connectivity to one or more of the peripheral devices that may include a wireless video display device **142**, a wireless keyboard **144**, a wireless mouse **150**, a wireless headset, a microphone, an audio headset, a wireless stylus **146**, and a wireless trackpad **148**, among other wireless peripheral devices used as input/output (I/O) devices **140**.

(24) The wireless interface adapter **126** may include any number of antennas **132** which may include any number of tunable antennas for use with the system and methods disclosed herein. Although FIG. **1** shows a single antenna **132**, the present specification contemplates that the number of antennas **132** may include more or less of the number of individual antennas shown in FIG. **1**. Additional antenna system modification circuitry (not shown) may also be included with the wireless interface adapter **126** to implement coexistence control measures via an antenna controller in various embodiments of the present disclosure.

(25) In some aspects of the present disclosure, the wireless interface adapter **126** may operate two or more wireless links. In an embodiment, the wireless interface adapter **126** may operate a Bluetooth wireless link using a Bluetooth wireless or Bluetooth extended protocols. In an embodiment, the Bluetooth wireless protocol may operate at frequencies between 2.402 to 2.48 GHz. Other Bluetooth operating frequencies such as Bluetooth extended frequencies at 6 GHz are also contemplated in the presented description. In an embodiment, a Bluetooth wireless link may be used to wirelessly couple the input/output devices operatively and wirelessly including the mouse **150**, keyboard **144**, stylus **146**, trackpad **148**, and/or video display device **142** to the bus **116** in order for these devices to operate wirelessly with the information handling system **100**. In a further aspect, the wireless interface adapter **126** may operate the two or more wireless links with a single, shared communication frequency band such as with the 5G standard relating to unlicensed wireless spectrum for small cell 5G operation or for unlicensed Wi-Fi WLAN operation in an example aspect. For example, a 2.4 GHz/2.5 GHz or 5 GHz wireless communication frequency bands may be apportioned under the 5G standards for communication on either small cell WWAN wireless link operation or Wi-Fi WLAN operation. In some embodiments, the shared, wireless communication band may be transmitted through one or a plurality of antennas **132** may be capable of operating at a variety of frequency bands. In an embodiment described herein, the shared, wireless communication band may be transmitted through a plurality of antennas used to operate in an N×N MIMO array configuration where multiple antennas **132** are used to exploit multipath propagation which may be any variable N. For example, N may equal 2, 3, or 4 to be 2×2, 3×3, or 4×4 MIMO operation in some embodiments. Other communication frequency bands, channels, and transception arrangements are contemplated for use with the embodiments of the present disclosure as well and the present specification contemplates the use of a variety of communication frequency bands.

(26) The wireless interface adapter **126** may operate in accordance with any wireless data communication standards. To communicate with a wireless local area network, standards including IEEE 802.11 WLAN standards (e.g., IEEE 802.11ax-2021 (Wi-Fi 6E, 6 GHz)), IEEE 802.15 WPAN standards, WWAN such as 3GPP or 3GPP2, Bluetooth standards, or similar wireless standards may be used. Wireless interface adapter **126** may connect to any combination of macro-cellular wireless connections including 2G, 2.5G, 3G, 4G, 5G or the like from one or more service providers. Utilization of radio frequency communication bands according to several example embodiments of the present disclosure may include bands used with the WLAN standards and WWAN carriers which may operate in both licensed and unlicensed spectrums. For example, both WLAN and WWAN may use the Unlicensed National Information Infrastructure (U-NII) band which typically operates in the ~5 MHz frequency band such as 802.11a/h/j/n/ac/ax (e.g., center frequencies between 5.170-7.125 GHz). WLAN, for example, may operate at a 2.4 GHz band, 5 GHz band, and/or a 6 GHz band according to, for example, Wi-Fi, Wi-Fi 6, or Wi-Fi 6E standards. WWAN may operate in a number of bands, some of which are proprietary but may include a wireless communication frequency band. For example, low-band 5G may operate at frequencies similar to 4G standards at 600-850 MHz. Mid-band 5G may operate at frequencies between 2.5 and 3.7 GHz. Additionally, high-band 5G frequencies may operate at 25 to 39 GHz and even higher. In additional examples, WWAN carrier licensed bands may operate at the new radio frequency range 1 (NRFR1), NRFR2, bands, and other known bands. Each of these frequencies used to communicate over the network **134** may be based on the radio access network (RAN) standards that implement, for example, eNodeB or gNodeB hardware connected to mobile phone networks (e.g., cellular networks) used to communicate with the information handling system **100**. In the example embodiment, the information handling system **100** may also include both unlicensed wireless RF communication capabilities as well as licensed wireless RF communication capabilities. For example, licensed wireless RF communication capabilities may be available via a subscriber carrier wireless service operating the cellular networks. With the licensed wireless RF

communication capability, a WWAN RF front end (e.g., antenna front end **130** circuits) of the information handling system **100** may operate on a licensed WWAN wireless radio with authorization for subscriber access to a wireless service provider on a carrier licensed frequency band.

(27) In other aspects, the information handling system **100** operating as a mobile information handling system may operate a plurality of wireless interface adapters **126** for concurrent radio operation in one or more wireless communication bands. The plurality of wireless interface adapters **126** may further share a wireless communication band or operate in nearby wireless communication bands in some embodiments. Further, harmonics and other effects may impact wireless link operation when a plurality of wireless links are operating concurrently as in some of the presently described embodiments.

(28) The wireless interface adapter **126** can represent an add-in card, wireless network interface module that is integrated with a main board of the information handling system **100** or integrated with another wireless network interface capability, or any combination thereof. In an embodiment the wireless interface adapter **126** may include one or more radio frequency subsystems including transmitters and wireless controllers for connecting via a multitude of wireless links. In an example embodiment, an information handling system **100** may have an antenna system transmitter for Bluetooth, 5G small cell WWAN, or Wi-Fi WLAN connectivity and one or more additional antenna system transmitters for macro-cellular communication. The RF subsystems and radios **128** and include wireless controllers to manage authentication, connectivity, communications, power levels for transmission, buffering, error correction, baseband processing, and other functions of the wireless interface adapter **126**.

(29) The information handling system **100** may be operatively coupled to an interim edge storage management system **154** via the network **134** using the wireless interface adapter **126**, radio **128**, and antenna **132** described herein. The interim edge storage management system **154** may be any device or devices, such as an on-premises server system, that process data for the execution of a storage orchestrator **156**. In an embodiment, the interim edge storage management system **154** may include a processing device (e.g., CPU, GPU, microcontroller, or other processing resource) that processes data including computer readable program code of the storage orchestrator **156**.

(30) In an embodiment, the interim edge storage management system **154** may receive authentication credentials from a plurality of client devices operatively coupled to the interim edge storage management system **154** such as the information handling system **100** described herein. The authentication credentials may include a username, a password, and/or other identification data that indicates that the user is allowed to access the interim edge storage management system **154** and its services. In an embodiment, the interim edge storage management system **154** may map each user to each of the plurality of network file hosting systems and one or more edge storage devices **159** to which the interim edge storage management system **154** allows the client devices to upload share file and other data to. In an embodiment, the interim edge storage management system **154** includes a storage orchestrator **156** that does this mapping as well as manage data storage of each of the client devices at the network file hosting systems **158** or one or more edge storage devices **159**. This storage orchestrator **156** may cause one or more APIs to be executed by the processor of the interim edge storage management system **154** for the storage orchestrator **156** to communicate with one or more network file hosting systems **158** or one or more edge storage devices **159**. Example network file hosting systems **158** may include, for example, Microsoft Azure® developed by Microsoft®, Google Drive® developed by Google®, Amazon Simple Storage Service @developed by Amazon®, and Microsoft OneDrive® developed by Microsoft®, among other network file hosting systems **158**.

(31) In an embodiment, the storage orchestrator **156** of the interim edge storage management system **154** may be responsible for moving files from one or more client devices of the enterprise environment. These files may be moved from an edge storage device **159** such as a virtual network-

attached edge storage device associated with the interim edge storage management system **154** to one or more of the network file hosting systems **158** or visa-versa described herein. In an embodiment, the storage orchestrator **156** may determine which files to be stored on a network file hosting system **158** or at the edge storage device **159** based on one or more active directory synchronization files received from a cloud computing management console **160**. These active directory synchronization files may be generated by a user of a client device, a manager of one or more client devices, or another IT administrator. In an embodiment, the IT administrator may provide client device-specific active directory synchronization files describing how files from a specific client device is to be saved, or not, to the network file hosting systems **158** or to the edge storage device **159**. In an embodiment, the IT administrator may provide enterprise-wide active directory synchronization files describing how files from each of the client devices within an enterprise or otherwise operatively coupled to the interim edge storage management system **154** are to be saved, or not, at one or more the network file hosting systems **158** or at an edge storage device **159**. In an embodiment, the active directory synchronization files may describe that most recently used (MRU) files that were accessed, updated, saved, or otherwise manipulated by a client device are to be stored on the edge storage device **159** such as a virtual network-attached edge storage device associated with the interim edge storage management system **154** in an embodiment. As the data storage capacities of the network file hosting systems **158** are reached (e.g., 5 TB in Google Drive), those files that are to be saved on the network file hosting systems **158** may be saved in a first-in-first-out process by deleting a file from a network file hosting system **158** before saving a new file to a network file hosting system **158** or some portion stored on an edge storage device **159**. It is appreciated that any other type of active directory synchronization files may be received and executed by the storage orchestrator **156** in order to determine which files on the client devices (e.g., information handling system **100**) are to be stored on a virtual network-attached edge storage device **159** associated with the interim edge storage management system **154** and/or on one or more of the network file hosting systems **158**.

(32) In an embodiment, the storage orchestrator **156** of the interim edge storage management system **154** may provide a snapshot of backups of client device files. In an embodiment, this snapshot may be descriptive of the state of a data storage device at a particular time including what files are stored where, when they were stored, and the file sizes of these files. This snapshot may also include data descriptive of data storage space left to store additional client device files in the virtual network-attached edge storage device **159** or among network file hosting systems **158**.

(33) The interim edge storage management system **154** may also include a virtual machine that maintains a local active directory. In an embodiment, the virtual machine may be operatively coupled to the virtual network-attached edge storage device **159**. The local active directory of the virtual machine may maintain a directory of client device files stored on the virtual network-attached edge storage device **159** in an embodiment. In an embodiment, the virtual machine may also receive the active directory synchronization files from the cloud computing management console **160** via a cloud active directory in order to pass those active directory synchronization files onto the storage orchestrator **156** as described herein.

(34) During operation, the information handling system **100** as one client device operatively coupled to the interim edge storage management system **154** may boot up and initiate communication with the interim edge storage management system **154**. In an embodiment, the information handling system **100** may provide user authentication data such as a username and password to authenticate the information handling system **100** to interface with the interim edge storage management system **154**. Because the interim edge storage management system **154** may serve as an edge storage device for multiple information handling systems **100**, the data and files stored on the interim edge storage management system **154** (e.g., a virtual network-attached edge storage device **159**) may be proprietary and accessible to authenticated and permitted users only.

(35) Once authenticated, the information handling system **100** may determine whether the storage

orchestrator **156** is available to manage data storage of the client device at the network file hosting system. Where the storage orchestrator **156** is not available to the user, data files may be stored on the local storage device of the information handling system. The storage orchestrator **156** may not be available to the user based on a number of reasons including, for example, lack of authorization to use the storage orchestrator **156**, updating of the computer readable program code executed by the storage orchestrator **156**, among other reasons. Where and when the storage orchestrator **156** is available to the user, that data and those files to be stored on the virtual network-attached edge storage device **159** associated with the interim edge storage management system **154** may be stored so that low data access latency may be maintained at the interim edge storage management system **154** by, for example, an enterprise or company for some data. For example, this low latency may provide data and files to a factory at, for example, an assembly line where access to that data and those files needs to be relatively faster than if these files were only stored in a remote network file hosting system **158** for example. However, this data and these files may also need to be accessible to a user via the information handling system **100** when the user has moved away from the enterprise. The storage orchestrator **156** may execute computer readable program code to determine if and when this data and these files are to be stored in one or more remote network file hosting systems **158** as well.

(36) During this time, a cloud computing management console **160** may be accessed by an IT administrator. This cloud computing management console **160** may be implemented to provide active directory synchronization files to the storage orchestrator **156**. The active directory synchronization files describe how the storage orchestrator manages data storage for the client device. In an example embodiment, the active directory synchronization files may provide directions to the storage orchestrator **156** so that certain files and data may be saved to certain network file hosting systems **158** based on a number of criteria. For example, the files may or may not be saved based on the type of files to be saved, the type of user of the information handling system **100** (e.g., corporation executive), and the remaining storage capacities of the network file hosting systems **158**, among other factors.

(37) The cloud computing management console **160** may synchronize the active directory synchronization files with the storage orchestrator **156** at any time and updates these active directory synchronization files when they are changed by the IT administrator or on a scheduled basis. In one embodiment, the active directory synchronization files may be updated every 7 days. Further, the interim edge storage management system **154** and storage orchestrator **156** may monitor and determine when remote network file hosting systems **158** approach or reach one or more threshold levels of storage capacity. Such thresholds may include when a remote network file hosting system **158** reaches a level where expansion of capacity on that remote network file hosting system **158** is required and available with no additional cost or registration or when expansion requires added cost and additional submission of subscription or authentication data to expand. With the cloud computing management console **160**, the IT administrator may be notified of approaching capacity limits in one embodiment. In another embodiment, the interim edge storage management system **154** and storage orchestrator **156** may operate to establish or utilize another API for setup and access to an added remote network file hosting system **158** to add capacity for the user or an enterprise. This addition of a new remote network file hosting system **158** to the existing one or more available may occur automatically with the interim edge storage management system **154** and storage orchestrator **156** in some embodiments. Then tracking stored files and updating file directories for between the new remote network file hosting system **158** and the edge storage device **159** during operation may be edge synched seamlessly and coordinated among plural remote network file hosting systems **158** potentially accessed by a client device **100**.

(38) The user operating the information handling system **100** may have a variety of methods of storing data. For example, the user may access data locally on the storage devices of the information handling system **100**. In an embodiment where specific data or files do not exist

locally, the information handling system **100** may, via the radio **128** and antenna **132**, may access the network **134** search for the requested data or files at the virtual network-attached edge storage device **159** of the interim edge storage management system **154**. Where the files are located on the virtual network-attached edge storage device **159**, the files may be accessed, again, in a fashion that has lower latency access than if that data or those files were stored on a remote network file hosting system **158**. In order to determine whether the data or files are stored on the virtual network-attached edge storage device **159** the information handling system **100** may request this information at the storage orchestrator **156** on the interim edge storage management system **154** which includes a directory of what data and files are stored and where those files are stored.

(39) The storage orchestrator **156** may store data and files on the virtual network-attached edge storage device **159** and/or the network file hosting systems **158** based on when that data and those files were last accessed. Where the user has accessed a file at the virtual network-attached edge storage device **159** or locally on the information handling system **100**, for example, the storage orchestrator **156** may be notified of this and, based on the instructions from the cloud computing management console **160** via the active directory synchronization files, save the MRU files to the virtual network-attached edge storage device **159** so that these files may be accessed by the information handling system **100** with low latency. In an embodiment, the active directory synchronization files may indicate that if the data or files were accessed at the network file hosting systems **158** or locally on the information handling system **100** within the last seven days, these files are saved to the virtual network-attached edge storage device **159** of the interim edge storage management system **154** so that this data and these files may be accessed quicker than if they were stored on the remote network file hosting system **158** in an embodiment. Where the files are determined to not exist on the virtual network-attached edge storage device **159** by the storage orchestrator **156**, the storage orchestrator **156** may access one or more file directories of the remote network file hosting systems **158** and determine where the user-requested data or files are being stored. In an embodiment, this data or files, when located, may be moved to the virtual network-attached edge storage device **159** of the interim edge storage management system **154** as a result of them being requested, accessed, updated, saved, or otherwise manipulated by a client device within a specific time period (e.g., within the last 24 hours). Where the data and files maintained on the virtual network-attached edge storage device **159** have not been accessed during an access time period threshold (e.g., 24 hours), those files may be replaced with any new MRU files using, for example, a first-in-first-out saving protocol. This may limit the amount of storage in the active directory synchronization files while still providing space for those files that have been most recently accessed, updated, saved, or otherwise manipulated by any given user of an information handling system **100**. This also allows certain data and files to be stored on the network file hosting systems **158** when they are not repeatedly accessed and do not need to be maintained on the virtual network-attached edge storage device of the interim edge storage management system **154**.

(40) The interim edge storage management system **154** described herein provides unlimited data storage for the enterprise of client devices without depending on the network file hosting system **158** for these data storage needs. Further, by maintaining the data on the virtual network-attached edge storage device **159** of the interim edge storage management system **154** via execution of the storage orchestrator **156** as described herein, the data and files may be more secure and more easily managed by an IT administrator. Still further, where needed, any existing subscriptions and access to the network file hosting systems **158** is maintained and the user or the IT administrator may not need to worry about storage limitations on those remote network file hosting systems **158**. In an embodiment, because the storage orchestrator **156** executes APIs that automatically provide access to one or more network file hosting systems **158**, the user is made unaware of where the files are stored. Still further, because the network file hosting systems **158** each have their own service costs and because each network file hosting system **158** may be under a service contract that includes a storage limit, the storage orchestrator **156** controls, seamlessly, the accessing and storage of any

given file on behalf of any client information handling system operatively coupled to the interim edge storage management system **154**. This access to the existing network file hosting systems **158** and, potentially, added network file hosting systems **158**, is maintained via execution of the one or more APIs at the storage orchestrator **156** and is used to access the individual network file hosting systems **158** in the cloud. Because the low latency of accessing the MRU data and files at the virtual network-attached edge storage device **159**, productivity and efficiency in operations may be increased during operations at the enterprise.

(41) The information handling system **100** can include one or more set of instructions **110** that can be executed to cause the computer system to perform any one or more of the methods or computer-based functions disclosed herein. For example, instructions **110** may execute various software applications, software agents, or other aspects or components. Various software modules comprising application instructions **110** may be coordinated by an operating system (OS) **114**, and/or via an application programming interface (API). An example OS **114** may include Windows®, Android®, and other OS types known in the art. Example APIs may include Win 32, Core Java API, or Android APIs.

(42) The disk drive unit **118** and may include a computer-readable medium **108** in which one or more sets of instructions **110** such as software can be embedded to be executed by the processor **102** or other processing devices such as a GPU **152** to perform the processes described herein. Similarly, main memory **104** and static memory **106** may also contain a computer-readable medium for storage of one or more sets of instructions, parameters, or profiles **110** described herein. The disk drive unit **118** or static memory **106** also contain space for data storage. Further, the instructions **110** may embody one or more of the methods as described herein. In a particular embodiment, the instructions, parameters, and profiles **110** may reside completely, or at least partially, within the main memory **104**, the static memory **106**, and/or within the disk drive **118** during execution by the processor **102** or GPU **152** of information handling system **100**. The main memory **104**, GPU **152**, and the processor **102** also may include computer-readable media.

(43) Main memory **104** or other memory of the embodiments described herein may contain computer-readable medium (not shown), such as RAM in an example embodiment. An example of main memory **104** includes random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof. Static memory **106** may contain computer-readable medium (not shown), such as NOR or NAND flash memory in some example embodiments. The applications and associated APIs described herein, for example, may be stored in static memory **106** or on the drive unit **118** that may include access to a computer-readable medium **108** such as a magnetic disk or flash memory in an example embodiment. While the computer-readable medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(44) In an embodiment, the information handling system **100** may further include a power management unit (PMU) **120** (a.k.a. a power supply unit (PSU)). The PMU **120** may manage the power provided to the components of the information handling system **100** such as the processor **102**, a cooling system, one or more drive units **118**, the GPU **152**, a video/graphic display device **142** or other input/output devices **140** such as the stylus **146**, a mouse **150**, a keyboard **144**, and a trackpad **148** and other components that may require power when a power button has been actuated by a user. In an embodiment, the PMU **120** may monitor power levels and be electrically coupled, either wired or wirelessly, to the information handling system **100** to provide this power and coupled to bus **116** to provide or receive data or instructions. The PMU **120** may regulate power

from a power source such as a battery **122** or A/C power adapter **124**. In an embodiment, the battery **122** may be charged via the A/C power adapter **124** and provide power to the components of the information handling system **100** via a wired connections as applicable, or when A/C power from the A/C power adapter **124** is removed.

(45) In a particular non-limiting, exemplary embodiment, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes or other storage device to store information received via carrier wave signals such as a signal communicated over a transmission medium. Furthermore, a computer readable medium can store information received from distributed network resources such as from a cloud-based environment. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

(46) In other embodiments, dedicated hardware implementations such as application specific integrated circuits (ASICs), programmable logic arrays and other hardware devices can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments can broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

(47) When referred to as a “system”, a “device,” a “module,” a “controller,” or the like, the embodiments described herein can be configured as hardware. For example, a portion of an information handling system device may be hardware such as, for example, an integrated circuit (such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a structured ASIC, or a device embedded on a larger chip), a card (such as a Peripheral Component Interface (PCI) card, a PCI-express card, a Personal Computer Memory Card International Association (PCMCIA) card, or other such expansion card), or a system (such as a motherboard, a system-on-a-chip (SoC), or a stand-alone device). The system, device, controller, or module can include software, including firmware embedded at a device, such as an Intel® Core class processor, ARM® brand processors, Qualcomm® Snapdragon processors, or other processors and chipsets, or other such device, or software capable of operating a relevant environment of the information handling system. The system, device, controller, or module can also include a combination of the foregoing examples of hardware or software. Note that an information handling system can include an integrated circuit or a board-level product having portions thereof that can also be any combination of hardware and software. Devices, modules, resources, controllers, or programs that are in communication with one another need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices, modules, resources, controllers, or programs that are in communication with one another can communicate directly or indirectly through one or more intermediaries.

(48) FIG. 2 illustrates a network **200** that can include one or more information handling systems **210**, **212**, **214**. The information handling systems **210**, **212**, **214** shown in FIG. 2 may be similar to the information handling system **100** and described in connection with FIG. 1, respectively. In an embodiment, the information handling systems **210**, **212**, **214** may be operatively coupled to an interim edge storage management system **254**. In an embodiment, the interim edge storage

management system **254** may be part of an edge device and may include a virtual network attached edge storage device **259** on part of the edge device or attached to the edge device that is placed between the information handling systems **210, 212, 214** (e.g., client devices) and one or more network file hosting systems **258**. The macro-cellular network **250** and/or wireless local network **240** may be used by the interim edge storage management system **254** to be operatively coupled to the interim edge storage management system **254**. In an embodiment, a cloud computing management console **261** may also be operatively coupled to the wireless local network **240** and/or macro-cellular network **250**. The cloud computing management console **160**, in an embodiment, may be operated by one of the information handling systems **210, 212, 214** or a dedicated computing device.

(49) In a particular embodiment, network **200** includes networked mobile information handling systems **210, 212, 214**, wireless network access points, and multiple wireless connection link options. A variety of additional computing resources of network **200** may include client mobile information handling systems, data processing servers, network storage devices, local and wide area networks, or other resources as needed or desired. As partially depicted, information handling systems **210, 212, 214** may be a laptop computer, tablet computer, 360-degree convertible systems, wearable computing devices, or a smart phone device. These information handling systems **210, 212, 214**, may access a wireless local network **240**, or they may access a macro-cellular network **250**. For example, the wireless local network **240** may be the wireless local area network (WLAN), a wireless personal area network (WPAN), or a wireless wide area network (WWAN). In an example embodiment, LTE-LAA WWAN may operate with a small-cell WWAN wireless access point option.

(50) Since WPAN or Wi-Fi Direct Connection **248** and WWAN networks can functionally operate similar to WLANs, they may be considered as wireless local area networks (WLANs) for purposes herein. Components of a WLAN may be connected by wireline or Ethernet connections to a wider external network such as a voice and packet core **280**. For example, wireless network access points (e.g., **136**, FIG. **1**) or base stations (e.g., **138**, FIG. **1**) may be connected to a wireless network controller and an Ethernet switch. Wireless communications across wireless local network **240** may be via standard protocols such as IEEE 802.11 Wi-Fi, IEEE 802.11ad WiGig, IEEE 802.15 WPAN, IEEE 802.11ax-2021, (e.g., Wi-Fi 6 and 6E, 6 GHz technologies), or emerging 5G small cell WWAN communications such as gNodeB, eNodeB, or similar wireless network protocols and access points. Alternatively, other available wireless links within network **200** may include macro-cellular connections **250** via one or more service providers **260** and **270**. As described herein, a plurality of antennas may be operatively coupled to any of the macro-cellular connections **250** via one or more service providers **260** and **270** or to the wireless local area networks (WLANs) selectively based on the SAR data, RSSI data, configuration data, system operation and connection metrics, peripheral telemetry data, and antenna mounting locations (e.g., spatial locations of antennas within the information handling system) associated with each information handling systems **210, 212, 214** as described herein. Service provider macro-cellular connections may include 2G standards such as GSM, 2.5G standards such as GSM EDGE and GPRS, 3G standards such as W-CDMA/UMTS and CDMA 2000, 4G standards, or emerging 5G standards including WiMAX, LTE, and LTE Advanced, LTE-LAA, small cell WWAN, and the like.

(51) Wireless local network **240** and macro-cellular network **250** may include a variety of licensed, unlicensed or shared communication frequency bands as well as a variety of wireless protocol technologies ranging from those operating in macrocells, small cells, picocells, or femtocells. As described herein, utilization of RF communication bands according to several example embodiments of the present disclosure may include bands used with the WLAN standards and WWAN carriers which may operate in both licensed and unlicensed spectrums. For example, both WLAN and WWAN may use the Unlicensed National Information Infrastructure (U-NII) band which typically operates in the ~5 MHz frequency band such as 802.11a/h/j/n/ac/ax (e.g., center

frequencies between 5.170-7.125 GHz). WLAN, for example, may operate at a 2.4 GHz band, 5 GHz band, and/or a 6 GHz band according to, for example, Wi-Fi, Wi-Fi 6, or Wi-Fi 6E standards. WWAN may operate in a number of bands, some of which are proprietary but may include a wireless communication frequency band. For example, low-band 5G may operate at frequencies similar to 4G standards at 600-850 MHz. Mid-band 5G may operate at frequencies between 2.5 and 3.7 GHz. Additionally, high-band 5G frequencies may operate at 25 to 39 GHz and even higher. In additional examples, WWAN carrier licensed bands may operate at the new radio frequency range 1 (NRFR1), NRFR2, bands, and other known bands. Each of these frequencies used to communicate over the network **240**, **250** may be based on the radio access network (RAN) standards that implement, for example, eNodeB or gNodeB hardware connected to mobile phone networks (e.g., cellular networks) used to communicate with the information handling systems **210**, **212**, **214**. In the example embodiment, mobile one or more information handling systems **210**, **220**, **230** may also include both unlicensed wireless RF communication capabilities as well as licensed wireless RF communication capabilities. For example, licensed wireless RF communication capabilities may be available via a subscriber carrier wireless service operating the cellular networks. With the licensed wireless RF communication capability, an WWAN RF front end of the information handling systems **210**, **212**, **214** may operate on a licensed WWAN wireless radio with authorization for subscriber access to a wireless service provider on a carrier licensed frequency band. WLAN such as Wi-Fi (e.g., Wi-Fi 6) may be unlicensed.

(52) As described herein, an interim edge storage management system **254** may operate with an associated virtual network attached edge storage device **259**. The interim edge storage management system **254** may be placed at an edge of the network, for example, on the premises at an enterprise in an embodiment. By placing the virtual network attached edge storage device **259** on the edge, the enterprise may access files stored on the virtual network attached edge storage device **259** of the interim edge storage management system **254** at a lower latency for quick access to those files.

(53) The voice and packet core network **280** shown in FIG. 2 may contain externally accessible computing resources and connect to a remote data center **286**. An external accessible computing resource at the remote data center **286** includes the one or more network file hosting systems **258** used by the interim edge storage management system **254** and storage orchestrator **256** to store, for example, data and files from the information handling systems **210**, **212**, **214** that are not the MRU files but are to be accessed by the information handling systems **210**, **212**, **214** remotely, for example. The voice and packet core network **280** may contain multiple intermediate web servers or other locations with accessible data (not shown). The voice and packet core network **280** may also connect to other wireless networks similar to the wireless local network **240** or the macro-cellular network **250** and additional mobile information handling systems such as **210**, **212**, **214**, or similar connected to those additional wireless networks. Connection **282** between the wireless network **240** and remote data center **286** or connection to other additional wireless networks may be via Ethernet or another similar connection to the world-wide-web, a WAN, a LAN, another WLAN, or other network structure. Such a connection **282** may be made via a WLAN access point/Ethernet switch to the external network and be a backhaul connection. The access point may be connected to one or more wireless access points in the WLAN before connecting directly to a mobile information handling system or may connect directly to one or more information handling systems **210**, **212**, **214**. Alternatively, mobile information handling systems **210**, **212**, **214** may connect to the external network via base station locations at service providers such as **260** and **270**. These service provider locations may be network connected via backhaul connectivity through the voice and packet core network **280**.

(54) Remote data centers **286** may include web servers or resources within a cloud environment that operate via the voice and packet core **280** or other wider internet connectivity. For example, remote data centers can include additional information handling systems, data processing servers, network storage devices (e.g., network file hosting systems **258**), local and wide area networks, or

other resources as needed or desired. In an embodiment, the remote data center **286** may include the machine data storage devices that store the computer code and instructions that define the remote network file hosting systems **258**.

(55) Although communication links **215**, **225**, and **235** are shown connecting wireless adapters of information handling systems **210**, **212**, **214** to wireless networks **240** or **250**, a variety of wireless links are contemplated. Wireless communication may link through a wireless access point (e.g., Wi-Fi), through unlicensed WWAN small cell base stations such as in network **240** or through a service provider tower and base stations such as that shown with service provider A **260** or service provider B **270** and in network **250**. In other aspects, mobile information handling systems **210**, **212**, **214** may communicate intra-device via inter-communication links **248** when one or more of the information handling systems **210**, **212**, **214** are set to act as an access point or even potentially an WWAN connection via small cell communication on licensed or unlicensed WWAN connections. For example, one of mobile information handling systems **210**, **212**, **214** may serve as a Wi-Fi hotspot in an embodiment. Concurrent wireless links to information handling systems **210**, **212**, **214** may be connected via any access points including other mobile information handling systems as illustrated in FIG. 2.

(56) FIG. 3 is a block diagram of a network edge server executing an interim edge storage management system **354** in an embodiment of the present disclosure. As described herein, the interim edge storage management system **354** may be on an edge device such as a server operatively coupled to one or more information handling systems **300-1**, **300-2**, **300-n**. The information handling systems **300-1**, **300-2**, **300-n** may each be client devices that use the interim edge storage management system **354** to determine where data and files accessed, updated, saved, or otherwise manipulated by the client devices are to be stored. As described herein in an embodiment, the interim edge storage management system **354** executes a storage orchestrator **356** that directs the storage of files and data to one of a virtual network-attached edge storage device **359** or one or more network file hosting systems **358-1**, **358-2**, **358-n** based on active directory synchronization files.

(57) The interim edge storage management system **354** may periodically receive the active directory synchronization files from a cloud computing management console **360** via a cloud active directory **366**. The cloud computing management console **360** may be used by an IT administrator **370** to create and push the active directory synchronization files that describe how the storage orchestrator **356** manages data storage for the client device. For example, the active directory synchronization files may indicate which files accessed by the information handling systems **300-1**, **300-2**, **300-n** are allowed to store data or files on the virtual network-attached edge storage device **359** and which of these data and files qualify as MRU data and files. In an embodiment, the cloud active directory **366** may authenticate and authorize all users operating an information handling system **300-1**, **300-2**, **300-n** and computers in a domain-type network, assigning and enforcing security policies for all information handling systems **300-1**, **300-2**, **300-n**, and installing or updating software, among other tasks apart from passing the active directory synchronization files to the interim edge storage management system **354**.

(58) The interim edge storage management system **354**, in an embodiment, includes a virtual machine **368** that includes an edge local active directory **368**. The virtual machine **364** may receive the active directory synchronization files from the cloud active directory **366** and maintain those with active directory synchronization for files stored on the virtual network-attached edge storage device **359** for access to the active directory synchronization files by the storage orchestrator **356**. The edge local active directory **368** may maintain data descriptive of those information handling systems **300-1**, **300-2**, **300-n** and their users allowed to and currently accessing and operatively coupled to the interim edge storage management system **354** as well as a directory of data and files stored on the virtual network-attached edge storage device **359**. The virtual machine **364** may, in an embodiment, assign processing resources associated with the interim edge storage management

system **354** to be used by the storage orchestrator **356** to manage the data and files accessed, updated, saved, or otherwise manipulated by a client device (e.g., information handling systems **300-1**, **300-2**, **300-n**).

(59) As described herein, the storage orchestrator **356** may trigger an edge synch process used to determine whether data or files have been accessed, updated, saved, or otherwise manipulated by an information handling system **300-1**, **300-2**, **300-n** within an access time period threshold (e.g., within the last 24 hours, last 7 days, or other access time period threshold). Where the data or files have not been accessed by any of the network file hosting systems **358-1**, **358-2**, **358-n** within that access time period threshold, the data or files may be saved to a network file hosting system **358-1**, **358-2**, **358-n**. Where the data or files have been accessed by any of the information handling systems **300-1**, **300-2**, **300-n** within that access time period threshold, the data or files may be saved to a virtual network-attached edge storage device **359** at the interim edge storage management system **354** to provide low latency access to those files while not recopying local storage and memory on the client information handling systems **300-1**, **300-2**, **300-n**. Further, use of virtual network-attached edge devices **362** further provides low-latency for shared files within the access time period threshold among files shared with a plurality of client information handling systems **300-1**, **300-2**, **300-n**. Further, in an embodiment, the storage orchestrator **356** manages files within the virtual network-attached edge storage device **359** that are to be saved therein. In an example embodiment, the storage orchestrator **356** may determine if and when the virtual network-attached edge storage device **359** capacity is full and add and remove the MRU files in a first-in-first-out process that, regardless of the access time period threshold, saves those files that were first saved in the virtual network-attached edge storage device **359** to a network file hosting system **358-1**, **358-2**, **358-n**.

(60) As described herein, the storage orchestrator **356** may cause one or more APIs to be executed by the processor of the interim edge storage management system **354** for the storage orchestrator **356** to communicate with one or more network file hosting systems **358-1**, **358-2**, **358-n** or virtual network-attached edge storage device **359**. Example network file hosting systems **358-1**, **358-2**, **358-n** may include, for example, Microsoft Azure® developed by Microsoft®, Google Drive® developed by Google®, Amazon Simple Storage Service® developed by Amazon®, and Microsoft OneDrive® developed by Microsoft®, among other network file hosting systems **358-1**, **358-2**, **358-n**. In an embodiment, because the storage orchestrator **356** executes APIs that automatically provide access to one or more network file hosting systems **358-1**, **358-2**, **358-n**, the user may be unaware of where the files are stored. The user has no need to provide specific credentials or authentication to access each or any of the one or more network file hosting systems **358-1**, **358-2**, **358-n**. In other words, this access to the network file hosting systems **358-1**, **358-2**, **358-n** or the virtual network-attached edge storage device **359** is completed without the user of the client information handling systems **300-1**, **300-2**, **300-n** needing to input usernames or login information to access the files. Access to any of these locations is seamlessly managed along with local storage at the client devices **300-1**, **300-2**, and **300-1** by the storage orchestrator **356** and with the edge local active directory **368** of the interim edge storage management system **354** and virtual machine **364**. Still further, because the network file hosting systems **358-1**, **358-2**, **358-n** each have their own service costs and because each network file hosting systems **358-1**, **358-2**, **358-n** under a service contract has a storage limit, the storage orchestrator **356** controls, seamlessly, the accessing and storage of any given file on behalf of any client information handling system operatively coupled to the interim edge storage management system **354**. During operation, additional access to other network file hosting systems **358-1**, **358-2**, **358-n** may be seamlessly provided if and when any given network file hosting system **358-1**, **358-2**, **358-n** becomes full.

(61) FIG. 4 is a block process diagram describing a method of operating an interim edge storage management system **454** according to an embodiment of the present disclosure. As described herein, an information handling system **400** may be used to initially authenticate the user with the

interim edge storage management system **454** via an authentication module **476** at the interim edge storage management system **454**. At this point, credentials **474** may be provided to the interim edge storage management system **454** by the information handling system **400** such as a username or password. The storage orchestrator **456** may also be provided with the user data for processing of data from or on behalf of the user of the information handling system **400**. Once authenticated by authentication module **476** at the interim edge storage management system **454**, the information handling system **400** may execute an agent **472** or other computer readable program code acting on behalf of the information handling system **400** to point to the storage at **478** and access data or files stored either on a virtual network-attached edge storage device **459** or on one or more of the network file hosting systems **458**.

(62) In an embodiment, the interim edge storage management system **454** may periodically receive the active directory synchronization files from a cloud computing management console **460** via a cloud active directory. The cloud computing management console **460** may be used by an IT administrator **470** to configure user policies and create and push the active directory synchronization files in a user policy configuration **488** that describe how the storage orchestrator **456** manages data storage for the client device **400**. For example, the active directory synchronization files may indicate which files accessed by the information handling systems **400** are allowed to store data or files **486** on the virtual network-attached edge storage device **459** and which of these data and files qualify as MRU data and files **482**. In an embodiment, the cloud active directory configuration **480** may authenticate and authorize all users operating an information handling system **400** and computers in a domain-type network, assigning and enforcing security policies for all information handling systems **400**, and installing or updating software, among other tasks apart from passing the active directory synchronization files for data and files synchronized to the cloud storage **484** to the interim edge storage management system **454**.

(63) The cloud computing management console **460** may push the active directory synchronization files for configuration **480** to the storage orchestrator **456**. Those files that are MRU files may be stored in the virtual network-attached edge storage device **459** associated with the interim edge storage management system **454** per the active directory synchronization files received by the storage orchestrator **456**. Those files that are deemed to not be recently used may be synchronized and stored in one or more network file hosting systems **458** in the cloud storage at **484** per the active directory synchronization files received by the storage orchestrator **456**.

(64) As described herein, the user may cause the information handling system **400** to request a file. This file may be located on the information handling system **400**, at the virtual network-attached edge storage device **459**, and/or at the network file hosting system **458**. The agent **472** may interface with the storage orchestrator **456** of the interim edge storage management system **454** in order to point to the storage at **478** where a copy of this file is located. This process is completed seamlessly without the user of the information handling system knowing that the agent **472** is accessing the file from the local storage of the information handling system **400**, from the virtual network-attached edge storage device **459**, or from the one or more network file hosting systems **458**. Where the file is located on one or more network file hosting systems **458**, the storage orchestrator **456** may cause one or more APIs to be executed by the processor of the interim edge storage management system **454** for the storage orchestrator **456** to communicate with one or more network file hosting systems **458** or virtual network-attached edge storage device **459**.

(65) FIG. 5 is a flow diagram of a method **500** of an information handling system operatively coupled to an edge device executing an interim edge storage management system according to an embodiment of the present disclosure. As described herein, the information handling system may be one of many information handling systems operatively coupled to the interim edge storage management system and the processes described in FIGS. 5 through 7 may be applicable to any number of information handling systems.

(66) The method **500** may begin with booting of the information handling system at block **505**. This booting process at block **505** may include executing a BIOS and OS on the information handling system. During the booting process, the processing device may cause a software agent associated with initiating a communication with the interim edge storage management system. This startup application may be represented as an icon on an application tray at the video display device for the user to access authentication settings. Upon startup of the software agent, the user may be requested, via the video display device, to enter in a username and password transmitted to the interim edge storage management system in order to gain access to the interim edge storage management system and the services provided by the storage orchestrator. The method **500** includes authenticating the information handling system with an authentication module of the interim edge storage management system at block **510** to verify the entered credentials to access the edge stored or remotely stored files. Any authentication of a user may be used including biometrics, embedded cryptographic keys or other authentication processes.

(67) The method **500** includes determining whether the storage orchestrator of the interim edge storage management system is available to manage data storage of the client device at plural storage locations including the virtual network-attached edge storage device and one or more the network file hosting systems at block **515**. Where the storage orchestrator is not available to the user, data files may be stored on the local storage device **521** at the client information handling system with the processing device of the client information handling system accessing this local data storage device **521** (e.g., main memory **104**, static memory **106**, computer readable medium **108**, etc. in FIG. **1**) at block **520**. The storage orchestrator may not be available to the user based on a number of reasons including, for example, lack of authorization to use the storage orchestrator and updating of the computer readable program code executed by the storage orchestrator, among other reasons. Because the storage orchestrator on the interim edge storage management system is not available in this example embodiment, the method **500** may end.

(68) Where the storage orchestrator of the interim edge storage management system is available at block **515**, the method **500** may include executing, with the processor of the information handling system, a software agent to point to the interim edge storage management system in order to allow the user to interact with the storage orchestrator thereon and access the services provided by the interim edge storage management system. As described herein, this may include the automatic storage of a user-created file and/or access of files stored by the storage orchestrator either on one of a plurality of network file hosting systems or a virtual network-attached edge storage device. It is appreciated, however, that via the execution of the software agent by the processor of the information handling system, the user is not made aware of whether the files being accessed or stored by the storage orchestrator are located on the network file hosting systems or on the virtual network-attached edge storage device. This relieves the user with having to know individual sign-in information the virtual network-attached edge storage device, the plurality of network file hosting systems, having to search for a specific file, or otherwise knowing what files are stored where. This allows the storage orchestrator of the interim edge storage management system to manage the storage and curation of those files without the involvement of the user.

(69) The method **500** includes, at block **525**, with determining whether a file is being created locally at the client information handling system. A file created locally may be saved on the virtual network-attached edge storage device associated with the interim edge storage management system so as to reduce the amount of data storage at the information handling system as well as allow the created files to be shared among a plurality of users on other client information handling systems also operatively coupled to and communicating with the interim edge storage management system. In this way, latency is reduced for the client information handling system and other client information handling systems for recent, locally created files.

(70) Where there are no files being created at block **525**, the method **500** may end. Where the file is being created at block **525**, the information handling system may access the interim edge storage

management system for storage of the created file on the virtual network-attached edge storage device at block **530**. In this example embodiment, because the file was created by the information handling system and is being saved by the storage orchestrator, this file qualifies as an MRU file and files within an access time period threshold such as an MRU file are saved onto the virtual network-attached edge storage device. The MRU files are saved on the virtual network-attached edge storage device in order to reduce the latency of access to these files.

(71) The method may include, at block **535**, with determining whether the information handling system is accessing any other files stored by the storage orchestrator online on, for example, the virtual network-attached edge storage device or on the one or more network file hosting systems. Where the user is not attempting to access one or more of these files at block **535**, the method **500** may continue with determine whether the information handling system is shut down or shutting down at block **537**. Where the information handling system is shutting down at block **537**, the method **500** may end. Where the information handling system is not shutting down at block **537**, the method **500** may continue with block **525** with monitoring for files being created by the client information handling system and whether the information handling system is requesting files located on a virtual network-attached edge storage device and/or one or more network file hosting systems.

(72) Where it is detected by the processor that the information handling system has requested access to one or more files online at block **535**, the method **500** may continue at block **540** with determining if the file exists on the local data storage device of the information handling system. Where those files are not located on the local data storage device of the information handling system, at block **540**, the method **500** may continue to block **545** with the information handling system sending a request to the storage orchestrator of the interim edge storage management system for these files. The storage orchestrator and the interim edge storage management system may access an active directory to locate the files at a virtual network attached edge storage device, for example if the file requested is an MRU from the requesting client device or another client device managed by the interim edge storage management system in an embodiment. Alternatively, the storage orchestrator may determine that the requested file is stored at one of a plurality of remote network file hosting systems and retrieve the file from the correct remote network file hosting system in another embodiment. This may cause the file to be added to virtual network attached edge storage device for future access and the active directory of the interim edge storage management system and the storage orchestrator to be updated. At this point, the method may proceed to block **546** with the user at a client device receiving those files stored online either from an edge storage device or from one or more remote network file hosting system with the location determined as described embodiments herein. Then the flow may proceed to block **537** to determine if the information handling system has shut down.

(73) Where the file requested by the user is saved locally on the information handling system as determined at block **540**, the method **500** includes providing those files to the information handling system directly at block **550**. Then the flow may proceed to block **537** to determine if the information handling system has shut down. In this way, one or more client information handling systems that are authenticated may seamlessly access files and data stored among local storage, virtual network-attached edge storage, or one or more network file hosting systems as determined and coordinate by the storage orchestrator according to embodiments herein.

(74) As before, at block **537**, the system may determine whether the information handling system is shut down or shutting down. Where the information handling system is shutting down at block **537**, the method **500** may end. Where the information handling system is not shutting down at block **537**, the method **500** may continue with block **525** with monitoring for files being created by the client information handling system and whether the information handling system is requesting files located on a virtual network-attached edge storage device and/or one or more network file hosting systems. FIG. **6** is a flow diagram of a method **600** of executing an interim edge storage

management system according to an embodiment of the present disclosure.

(75) FIG. 6 is from the perspective of an interim edge storage management system that is operatively coupled to one or more information handling systems and one or more network file hosting systems. The interim edge storage management system, in an embodiment, acts as an intermediary device that directs the storage of data and files to one of a virtual network-attached edge storage device or one or more network file hosting systems. The files saved may be created by one or more client information handling systems that are, at least, backed up on one of the virtual network-attached edge storage devices or the one or more network file hosting systems in an embodiment.

(76) The method **600** may include, at block **605** with receiving user authentication from one or more of the information handling systems operatively coupled to the interim edge storage management system. The user authentication may include a password, a username, or other login information. Any authentication of a user may be used including password or login credentials, biometrics, embedded cryptographic keys or other authentication processes. In an embodiment, the login or authentication information may allow a user to access the storage resources, at least, of the interim edge storage management system (e.g., virtual network-attached edge storage device) and one or more network file hosting systems without the user of the client device needing to manage access to these multiple storage sources. In an embodiment, the client information handling systems operatively coupled to this interim edge storage management system include an agent to automatically provide the authentication data and access the services provided by the interim edge storage management system. This authentication process is completed seamlessly without the user of the information handling system knowing that the agent is requesting, in the embodiments herein, the interim edge storage management system to access a file from the local storage of the information handling system, from the virtual network-attached edge storage device, or from the one or more network file hosting systems. Where the file is located on one or more network file hosting systems, a storage orchestrator of the interim edge storage management system may cause one or more APIs to be executed by the processor of the interim edge storage management system for the storage orchestrator to communicate, seamlessly, with one or more network file hosting systems **458** or virtual network-attached edge storage device. Further, the interim edge storage management system may manage file locations for availability to a plurality of client information handling systems that provide authentication information to allow seamless and efficient sharing of these files among the plurality of client information handling systems in some embodiments. For example, the plurality of client information handling systems may be managed by an enterprise which also administers the interim edge storage management system.

(77) The method **600** may include a determination as to whether the user is authorized to access the interim edge storage management system at block **610**. As described herein, the interim edge storage management system may direct the storage of proprietary information and access to these files and data may be limited to only those users, whether part of an enterprise or not, who are allowed to view the information. Where the user is not authorized at block **610**, the method may end. Where the user is authorized at block **610**, the method may continue to block **615**.

(78) In an embodiment, the method includes triggering an edge synch agent to synchronize files created at the information handling system at block **615**. The edge synch agent may be used to synchronize those files that have been created on the information handling system with those stored on the virtual network-attached edge storage device or the one or more network file hosting systems. In an embodiment, the edge synch agent may be computer readable program code that is executed by processing resources on the interim edge storage management system. In an embodiment, the edge synch agent may also request and receive active directory synchronization files from a cloud computing management console that instruct the storage orchestrator of the interim edge storage management system as to how and under what conditions the storage orchestrator stores data or files on the virtual network-attached edge storage device or one or more

network file hosting systems.

(79) Where the user is authorized at block **610** and the edge such agent has been triggered at block **615**, the method **600** may include determining whether the files are being created by the information handling system at block **620**. Where there are no files being created by the information handling system at block **620**, the method may continue to block **630** with determining whether the interim edge storage management system has received a request from a client information handling system to access online files (e.g., at the virtual network-attached edge storage device or on one or more network file hosting systems). This is described in more detail herein.

(80) Where it has been determined that the files were created at the client information handling system at block **620**, the synchronization of these created files may include, at block **625**, saving those files to a virtual network-attached edge storage device **659**. As described herein, because the file was created and is now being saved, the access time period threshold for the file to qualify as an MRU file has been met and the file is saved to the virtual network-attached edge storage device **659** of the interim edge storage management system. The edge synch agent will synchronize those files that have been created on the information handling system and determine location to store those files on the virtual network-attached edge storage device **659** or the one or more network file hosting systems **658**. In an embodiment, the edge synch agent will also update the active directory synchronization files from a cloud computing management console such that the storage orchestrator of the interim edge storage management system may track data or files are to be located on the virtual network-attached edge storage device **659** or one or more network file hosting systems **658**. The determination that a file is being created on a client information handling system allows the client information handling system or any other additional client information handling systems to access the created file at the virtual network attached edge storage device **549** by the interim edge storage management system instead of on a network file hosting system **658** thereby reducing the latency in retrieving that information. As an example, if the created file was an operation standard for an assembly line that is used to coordinate operations on an assembly line floor, the lower latency of retrieving this information will reduce costs associated with any downtime at the assembly line. In another example, if the file is an executable program describing how a self-driving car is to operate, the low latency may prevent damage to the car or injury to an occupant.

(81) The method **600** may continue, at block **630** with determining whether a request to access files has been made from a client information handling system or a plurality of client information handling systems. These files may include online or remote files that are stored on one or more network file hosting systems or a virtual network-attached edge storage device associated with the interim edge storage management system rather than locally on the client information handling system. Where a user is not accessing these online files at block **630**, the method **600** may utilize local storage or may determine files are unavailable and continue with block **675** with determining whether the interim edge storage management system is shut down or shutting down. Where the interim edge storage management system is shut down, the method **600** may end. Where the interim edge storage management system is not shut down, the method may continue to block **635** described herein.

(82) Where the client information handling system request is to access online or remotely-stored files, the user, in attempting to access these online or remote files, may not be made aware of where any given files is stored whether it is on the virtual network-attached edge storage device or the one or more network file hosting systems at block **630**. In an embodiment, the user may interface with a storage orchestrator of the interim edge storage management system that references a local active directory or cloud active directory to determine where these files are stored or copied. To the user, the file access may appear no differently than an access of a local file or there may be some indication when a file is online or remote, but there may be seamless access to those files regardless

of whether it is on the virtual network-attached edge storage device or the one or more network file hosting systems.

(83) Where a user is accessing these online files at block **630**, the method **600** may continue at block **635** with determining whether the files being requested by the user are located on the virtual network-attached edge storage device. This may be conducted with the storage orchestrator accessing a local active directory for the current files stored at the virtual network-attached edge storage device in some embodiments. Where the storage orchestrator determines that the requested files are stored on the virtual network-attached edge storage device, the method **600** continues to block **637** with accessing the virtual network-attached edge storage device at **659** for the user and directing the user to that file for access. The virtual network-attached edge storage device at **659** may be any, for example, server with memory storage capacity that is co-located at or near the operation of a client information handling system such as on premises of an enterprise or at a location within a localized geographic area for a nearby network access such as a building, campus, neighborhood, town, county, metropolitan area or similar that is closer than a remote data center located in a different metropolitan area, state or even country. Again, the agent associated with the client information handling system may interface seamlessly with this process such that the user is not aware of where the file is stored. Further, such accesses may update the local active file directory and reset the time period monitoring relative to an access time period threshold for MRU designation among other aspects of the synchronization and the interim edge storage management system in various embodiments. Upon accessing and retrieving the file from the virtual network-attached edge storage device **659**, the flow may proceed to block **675** to determine if the interim edge storage management system is shutting down. If so, the process may end. If not, the process may return to block **620** to monitor for additional files being created or modified at client information handling systems or for monitoring for file access requests, such as at **630**, from client information handling systems as described.

(84) Where the user-requested online file is not on the virtual network-attached edge storage device **659** at block **635**, the method **600** includes determining whether the online files are located on one or more of the network file hosting systems **658** at block **640**. Further, the storage orchestrator may access a cloud active directory that is synched with the local active directory to determine if and which network file hosting system **658** may have the stored file. Again, the one or more of the network file hosting systems **658** may be any, for example, subscription-based file hosting system that, in an embodiment, includes data storage limits. Example network file hosting systems **658** may include, for example, Microsoft Azure® developed by Microsoft®, Google Drive® developed by Google®, Amazon Simple Storage Service® developed by Amazon®, and Microsoft OneDrive® developed by Microsoft®, among other network file hosting systems. Such network file hosting systems may store data at a remote data center or centers located at further distances from the client information handling system, but may provide greater storage resources at each or among them. Where it is determined at block **640** that the user-requested files are not on one or more of the network file hosting systems, the method **600** may proceed with, in an embodiment, the storage orchestrator notifying the client device (e.g., an information handling system operated by the user) that the file is not found at block **645**. Where the storage orchestrator, via the cloud active directory, determines that the user-requested file is on one or more network file hosting systems and on which one it is stored at block **640**, the method continues to block **650** with accessing the network file hosting system **658** where the file is stored. Then the file may be retrieved by the interim edge storage management system and provided to the requesting client information handling system. At this point the edge synch agent and storage orchestrator may update the active local directory and the active cloud directory of the interim edge storage management system and store the retrieved file at the virtual network-attached edge storage device as it is a recent access in an embodiment. As described herein, the network file hosting system **658** may be one of a plurality of network file hosting systems where a single file is stored among them and/or backed up by

another network file hosting system or the virtual network-attached edge storage device. Upon accessing and retrieving the file from the remote network file hosting system **658**, the flow may proceed to block **675** to determine if the interim edge storage management system is shutting down. If so, the process may end. If not, the process may return to block **620** to monitor for additional files being created or modified at client information handling systems or for monitoring for file access requests, such as at **630**, from client information handling systems as described.

(85) In another aspect, interim edge storage management system the virtual network-attached edge storage device **659** operating at block **615** and determining authorization and synchronization of files with client information handling systems may also include monitoring function to monitor access frequency of files within the local active directory on the virtual network-attached edge storage device **659** or in the cloud active directory in synchronizing files with one or more network file hosting systems **658**. The virtual network-attached edge storage device **659** may have limited storage capacity relative to the network file hosting systems **658** and thus, storage at the edge storage device location may be limited as among one or more client information handling systems accessing this storage resource based upon recency of use for files, frequency of use of files, or criticality of file access speed based on file types or application context executing on the client information handling system or systems. Such a combination of factors are used to determine priority of those files to remain in the active local directory for storage at the virtual network-attached edge storage device **659** instead of the network file hosting systems **658**.

(86) In one embodiment, an access time-period threshold may determine the status of a file as an MRU file. The interim edge storage management system may determine or cull files that fall outside of the access time-period threshold from the local active directory and the virtual network-attached edge storage device **659** and transition those files in the cloud active directory and select storage at one of the network file hosting systems **658**. Further, if the virtual network-attached edge storage device **659** is nearing storage capacity, new files may be added while last in files may be transitioned to the network file hosting system **658** and the local active directory and cloud active directories updated accordingly for the storage orchestrator.

(87) In yet other embodiments, contextual priority may be attached to files with metadata from the client information handling system or via the interim edge storage management system. For example, certain types of files, such as archival files may not require low-latency access relative to files such as those used in automated manufacturing operations, self-driving or robotic self-operating applications, or other latency critical files during execution of applications by one or more client information handling systems. The contextual priority may further be adjustable by an IT administrator of the interim edge storage management system to provide certain software applications operating on client information handling system a high priority designation or others a low priority designation based on received metadata identifying the type of file to be stored among the virtual network-attached edge storage device **659** or the network file hosting systems **658** for particular contexts in an example embodiment. There may be a ranking system of application contexts for files stored and managed by the interim edge storage management system in an embodiment that may include two or more levels of ranking priority of contextual file types. This contextual priority may be an additional factor to the access time-period threshold measurement and assessment in some embodiments. Further, the interim edge storage management system may include a frequency of access measurement and use of this assessment by the interim edge storage management system to assign priority relative to files accessed infrequently of a access frequency time period as distinct from an access (recency) time period as discussed above. This frequency priority may be an additional factor to the access time-period threshold measurement and contextual priority factors in some embodiments.

(88) At block **660**, the storage orchestrator of the interim edge storage management system may concurrently determine whether the user has accessed this file on the network file hosting system within the access time period threshold (e.g., last 24 hours) and may weight this assessment in an

embodiment by contextual priority rating or by frequency of access ratings. As described herein, the storage orchestrator and the interim edge storage management system allows for low latency access to files that any given client device has accessed. Such low latency access may further be weighted toward high priority contextual file types or those that are frequency accessed. At block **670**, those files may be stored on a virtual network-attached edge storage device and a local active directory and the cloud active directory updated accordingly for the storage orchestrator.

(89) At block **665**, files that exceed an access time-period threshold may be transitioned to be stored on one or more network file hosting systems when it is determined that the files no longer qualify as MRU files (e.g., those files not accessed within the access time period threshold). In some embodiments, the access time-period threshold determination may be weighted with the additional contextual or frequency weighting factored in as well. Where it is determined at block **660** that the file has not been accessed within the access time period threshold, the method **600** may include at block **665** transitioning to store those files to a network file hosting system at **658**.

Where, at block **660**, it has been determined that the user has accessed this file within the access time period threshold (or any other threshold period of time based on the active directory synchronization files received by the storage orchestrator), the method **600** includes storing those files accessed by the user on the virtual network-attached edge storage device at **659**.

(90) After storage of any files within the virtual network-attached edge storage device at block **659** at block **670** or, at one or more of the network file hosting systems at block **665**, the method **600** may continue with block **675** with determining whether the interim edge storage management system is shut down or shutting down. Where the interim edge storage management system is shut down, the method **600** may end. Where the interim edge storage management system is not shut down, the method may continue to block **620** and proceed to monitor for files to be saved or accessed by client information handling system via the interim edge storage management system as described in embodiments herein.

(91) FIG. 7 is a flow diagram of a method **700** of executing one or more network file hosting systems with an interim edge storage management system according to an embodiment of the present disclosure. The method **700** may be descriptive of storing data and files on one or more network file hosting systems and may be from the perspective of a storage orchestrator according to information provided by a cloud active directory as well as synchronization and management of files between the network file hosting systems and the virtual network-attached edge storage device. In an aspect, the interim edge storage management system may provide a service to add one or more addition remote network file hosting systems when a storage threshold is approached at existing network file hosting systems in an embodiment. This may be done seamlessly according to embodiments herein with the interim edge storage management system establishing an API and authentication for access to a newly added network file hosting system for additional cloud storage capacity. Further, the interim edge storage management system may then provide coordination and synchronization of data files as between the network file hosting systems including the newly added network file hosting system and the virtual network-attached edge storage device as with other embodiments.

(92) The method **700** includes, at block **705**, the interim edge storage management system (e.g., the storage orchestrator) to update a file directory. The file directory may include a cloud active directory to define to the storage orchestrator of the interim edge storage management system what files are stored where among the one or more network file hosting systems. As described in additional previous embodiments herein, the interim edge storage management system may also maintain an active local directory as part of the file directory for synchronization and coordination for file storage at the virtual network-attached edge storage device. In an embodiment, a single copy of any given file may be saved in a virtual network-attached edge storage device and in one or more of the network file hosting systems or may be alternatively stored between these two storage resources. In one embodiment, the one or more network file hosting systems may serve as a back-

up location for files stored on the virtual network-attached edge storage device. The file directory, including the active local directory and the cloud active directory, may be updated periodically or when requested by the interim edge storage management system as indicated at block **705**.

(93) The method **700** may also include, at block **710**, accessing the one or more network file hosting systems with the storage orchestrator executing one or more APIs as described herein. Because the storage orchestrator executes APIs that automatically provide access to one or more network file hosting systems, the user of a client information handling system is unaware of where the files are stored since the interim edge storage management system handles coordination of this. Still further, because the network file hosting systems **758** each have their own service costs and because each network file hosting system **758** may be under a service contract that includes one or more storage limits that may invoke additional costs or additional contractual engagement upon reaching one or more storage thresholds, the interim edge storage management system may coordinate among plural network file hosting systems. Additionally, the interim edge storage management system may add new network file hosting systems **758** when certain storage limits are reached rather than enter into complicated storage expansion procedures or pay higher storage costs for a nearly full network file hosting system in embodiments herein. Thus, the interim edge storage management system with the storage orchestrator controls, seamlessly, the accessing and storage of any given file on behalf of any client information handling system operatively coupled to the interim edge storage management system as among the existing and the addition of new network file hosting systems as needed. Examples of a plurality of network file hosting systems managed by the interim edge storage management system may include, for example, Microsoft Azure® developed by Microsoft®, Google Drive® developed by Google®, Amazon Simple Storage Service® developed by Amazon®, and Microsoft OneDrive® developed by Microsoft®, among other network file hosting systems. As described herein, the storage orchestrator may direct the saving of files or data to any of these plurality of network file hosting systems **758**. In an embodiment, the storage orchestrator may have directed that certain files be placed within certain network file hosting systems **758** while other network file hosting systems **758** are used as backup sites for those files. Still further, in an example embodiment, the storage orchestrator may direct those files and data be saved among each of the network file hosting systems **758** operatively coupled to the storage orchestrator to maintain, roughly, even amounts of data for the enterprise. In an embodiment, the storage orchestrator of the interim edge storage management system may monitor for storage limits in each of the network file hosting systems.

(94) The method **700** includes at block **715** determining whether any of the network file hosting systems is at a threshold storage limit. As described herein, the interim edge storage management system may determine for each the network file hosting systems its customized data and file storage limit that is set, in some example embodiments, by a subscription agreement between the network file hosting system and the enterprise/client device. For example, customers of OneDrive® by Microsoft® may be limited to 1 terabyte (1 TB) per user. These customers may be able to increase the data storage (e.g., 5 TB) with an increased subscription cost. Further, customers may be able to increase data storage above 5 TB with a complicated change to the contractual terms of the data storage service. Any additional storage capacities may be limited to the users by administrators of the network file hosting system and may be granted on a case-by-case basis. These extra steps may be cumbersome to the user or an administrator and may increase the costs of operation for the user. Still further, latency issues may arise when a user is attempting to access the files stored on the network file hosting system.

(95) Where it is determined, at block **715**, that one or more of the network file hosting systems have not reached a threshold storage limit such that remote storage capacity is sufficient for the interim edge storage management system, the method may continue at block **720** with determining whether the interim edge storage management system is shut down or shutting down. Where the interim edge storage management system is shut down, the method **700** may end. Where the

interim edge storage management system is not shut down, the method may continue to block 725 as described herein. At block 715 and prior to block 720, the interim edge storage management system may allow for the storage orchestrator to save any files or data to the network file hosting system that are determined to not meet the requirements of being an MRU file.

(96) Where it is determined, at block 720, that any of the plurality of network file hosting systems are at a threshold storage limit, the method 700 includes mapping a user/client device storage to an alternative network file hosting system that has more storage space at block 725. This may include searching for a network file hosting system that has not yet reached its threshold data storage limits in an embodiment. This may assist in determining whether to add an additional network file hosting system or whether the currently engaged network file hosting systems, if more than one are engaged by the interim edge storage management system, are of sufficient capacity to accommodate the operations of interim edge storage management system in servicing the one or more client information handling systems.

(97) In an example embodiment where all accessible network file hosting system are approaching a threshold storage limit such as to incur additional storage costs or to trigger requirement of storage provider approvals to utilize even greater storage capacity at that service for a network file hosting system, the interim edge storage management system with the storage orchestrator may create a new account with an existing or a new network file hosting system in order to provide more storage capacity for those non-MRU files or back-ups of MUR files. In an embodiment, at block 730, the method 700 may include adding this alternative network file hosting system to the file directory by, for example, initiating a new subscription with a network file hosting system not yet operatively coupled to the storage orchestrator. Alternatively, this new subscription may include entering into an increased data storage limit with an existing network file hosting system operatively coupled to the storage orchestrator. With a newly added network file hosting system, the interim edge storage management system may establish an API and authentication credential with the newly added network file hosting system for access to storage located with that service in an embodiment. Further, the interim edge storage management system may add the newly added network file hosting system to the cloud active directory and the file directory of the storage orchestrator and being coordination and synchronization of file storage and access with the newly added network file hosting system as well as previous network file hosting systems and the virtual-network attached edge storage device according to the embodiments herein.

(98) Upon adding the new, alternative file hosting system at block 730, the method 700 may continue with determining whether the interim edge storage management system is shut down or shutting down at block 720. Where the interim edge storage management system is shut down, the method 700 may end. Where the interim edge storage management system is not shut down, the method may continue to block 710 as described herein.

(99) The blocks of the flow diagrams of FIGS. 5 through 7 or steps and aspects of the operation of the embodiments herein and discussed above need not be performed in any given or specified order. It is contemplated that additional blocks, steps, or functions may be added, some blocks, steps or functions may not be performed, blocks, steps, or functions may occur contemporaneously, and blocks, steps or functions from one flow diagram may be performed within another flow diagram.

(100) Devices, modules, resources, or programs that are in communication with one another need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices, modules, resources, or programs that are in communication with one another can communicate directly or indirectly through one or more intermediaries.

(101) Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be

included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

(102) The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover any and all such modifications, enhancements, and other embodiments that fall within the scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A method of operating an interim edge storage management system of a storage management information handling system operatively coupled with an edge storage device and a plurality of remote network file hosting systems comprising: accessing, with a processor executing code instructions of the interim edge storage management system, one or more storage policies, the storage policies being descriptive of how a plurality of data files used by a client device are to be stored on the edge storage device and on one or more of the plurality of remote network file hosting systems; with the processor, receiving credentials of the client device and authenticating the client device for use with the interim edge storage management system to access the edge storage device and the plurality of remote network file hosting systems managed by the interim edge storage management system; with the processor executing computer readable program code of a storage orchestrator, directing how the storage of a first set of existing data files accessed, updated, saved, or otherwise manipulated by the client device are selected to be stored among the edge storage device and the one or more of the plurality of remote network file hosting systems based on recency of usage of the first set of existing data files; with the processor executing computer readable program code of a storage orchestrator, manage access by the client device to the edge storage device and the plurality of remote network file hosting systems via an edge local active directory synched with a cloud active directory of the edge storage device and the plurality of remote network file hosting systems and selecting storage of the existing data files accessed, updated, saved, or otherwise manipulated by the client device from among the edge storage device and the plurality of remote network file hosting systems for the storage of one or more data files from a most recently used (MRU) data set to be stored on the edge storage device and one or more data files from a less recently used data set to be stored on the plurality of remote network file hosting systems, wherein the first MRU data set includes data files accessed, updated, saved, or otherwise manipulated by the client information handling system that are different from the second less recently used data set of the client information handling system; gathering information on storage capacity limits being reached at a first remote network file hosting system with a first hosting service of the plurality of remote network file hosting systems the storage orchestrator; and transitioning storage of a portion of the second less recently used data set of existing data files accessed, updated, saved, or otherwise manipulated by the client device to a second remote network file hosting system with a second hosting service of the plurality of remote network file hosting systems based on the gathered information of storage capacity limits.

2. The method of claim 1 further comprising: adding a new remote network file hosting system to the plurality of remote network file hosting systems based on storage capacity thresholds being reached at the first remote network file hosting system.

3. The method of claim 1 further comprising: managing, with the processor executing computer readable code instructions of the storage orchestrator, the storage of the first set of existing data files and a second set of recent data files from accessing, updating, saving, or otherwise

manipulating by the client device onto the edge storage device based on the usage recency of those first set of existing data files and the second set of recent data files and application of a application priority context to the first set of existing data files and the second set of recent data files pursuant to a storage policy for application priority for data files to be stored at the edge storage device.

4. The method of claim 1 further comprising: storing, with the processor executing computer readable code instructions of the storage orchestrator, a recent data file from the second set of recent data files from accessing, updating, saving, or otherwise manipulating by the client device within edge storage device and replacing the data file from the first set of existing data files in the edge storage device using a first-in-first-out data storage policy.

5. The method of claim 1 further comprising: automatically storing, synchronizing, and transitioning, with the processor executing computer readable code instructions of the storage orchestrator, data files from the first set of existing data files and the second set of recent data files stored from plural client devices operatively coupled to the interim edge storage management system of the storage management information handling system across the edge storage device and the plurality of remote network file hosting systems.

6. The method of claim 1 further comprising: executing, with the processor executing computer readable code instructions of the storage orchestrator, one or more application programming interfaces (APIs) to interface the storage orchestrator with the plurality of remote network file hosting systems from a plurality of hosting services, where a portion of the plurality of hosting services require different authentication and access requirements from other hosting services; and gathering data storage information related to the storage capacity limits of the plurality of remote network file hosting systems.

7. A storage management information handling system for providing storage management services to a client information handling system comprising: a processor; a memory device; a power management unit (PMU); the processor executing computer readable program code instructions of an interim edge storage management system, the processor of the storage management information handling system operatively coupled to the client information handling system, an edge storage device, and a plurality of remote network file hosting systems managed by the interim edge storage management system executing on the storage management information handling system; the processor executing computer readable program code instructions of the interim edge storage management system to receive credentials of the client information handling system and authenticate the client information handling system for use with the interim edge storage management system to access the edge storage device and the plurality of remote network file hosting systems managed by the interim edge storage management system; the processor executing computer readable program code instructions of the interim edge storage management system that includes a storage orchestrator to manage access by the client information handling system to select among the edge storage device and the plurality of remote network file hosting systems via an edge local active directory synched with a cloud active directory of edge storage devices and the plurality of remote network file hosting systems; and the processor executing computer readable program code instructions of the interim edge storage management system to select storing a first most recently used (MRU) data set of the client information handling system from data files most recently accessed, updated, saved, or otherwise manipulated by the client information handling system at the edge storage device and select storing a second less recently used data set of data files less recently accessed, updated, saved, or otherwise manipulated by the client information handling system at the plurality of remote network file hosting systems managed by the interim edge storage management system, wherein the first MRU data set includes data files accessed, updated, saved, or otherwise manipulated by the client information handling system that are different from the second less recently used data set of the client information handling system.

8. The storage management information handling system of claim 7 further comprising: the processor executing computer readable program code instructions of the storage orchestrator to

gather data storage information related to the storage of one or more of the data files from the first MRU dataset stored on the edge storage device or the storage of one or more of the data files of the second less recently used data set stored on the plurality of remote network file hosting systems; and the storage orchestrator to adjust the amount of data stored on the edge storage device to at least one of the plurality of remote network file hosting systems based on one or more storage policies by transitioning files from the edge storage device to the at least one remote network file hosting system based on assessment of the data files of the first MRU dataset under the one or more storage policies.

9. The storage management information handling system of claim 7 further comprising: the processor executing computer readable program code instructions of the storage orchestrator to gather data storage information related to the storage of one or more files of the second less recently used data set stored on the plurality of remote network file hosting systems; and the storage orchestrator to adjust the amount of data stored on the plurality of remote network file hosting systems based on one or more storage policies applied to the one or more files of the second less recently used data set by transitioning files from a first remote network file hosting system to a second remote network file hosting system based on storage capacity thresholds being reached at the first remote network file hosting system.

10. The storage management information handling system of claim 7 further comprising: the processor executing computer readable program code instructions of the storage orchestrator managing the storage and transitioning of the data file onto the edge storage device based on the detected recency of usage of the data file by accessing, updating, saving, or otherwise manipulating by the client information handling system and an application priority context storage policy applied to the data file from the first MRU dataset to be stored at the edge storage device.

11. The storage management information handling system of claim 7 further comprising: the processor executing computer readable program code instructions of the storage orchestrator managing the storage and transitioning of the file onto the edge storage device based on the recency of usage of the data file by accessing, updating, saving, or otherwise manipulating by the client information handling system and based on a frequency of access of the data file from the first MRU dataset by the client information handling system.

12. The storage management information handling system of claim 7, wherein the storage orchestrator transitions data files to the plurality of remote network file hosting systems when access to those data files falls outside of an access time-period threshold such that the data files are part of the second less recently used dataset accessed by the information handling system.

13. The storage management information handling system of claim 7 further comprising: the processor executing computer readable program code instructions of the storage orchestrator operatively coupled to the edge storage device storing and tracking data files of the from the first MRU dataset from the client information handling system for storage on the edge storage device and enforcing rules for actively transitioning other data files on the edge storage device to at least one remote network file hosting system based on one or more priority factors from storage policies.

14. The storage management information handling system of claim 7 further comprising: the processor executing one or more application programming interfaces (APIs) executable for the storage orchestrator to interface with the plurality of remote network file hosting systems that are diverse remote network file hosting systems having file directories and formatting developed by a plurality of network file hosting services.

15. A storage management information handling system for providing storage management services to a client device comprising: a processor; a memory device; a power management unit (PMU); and the processor executing computer readable program code of an interim edge storage management system, the processor of the storage management information handling system operatively coupled to the client device, an edge storage device, and a plurality of remote network file hosting systems managed by the interim edge storage management system to manage storage of

data files from the client device among the edge storage device and the plurality of remote network file hosting systems; the processor executing computer readable program code of the interim edge storage management system to receive credentials of the client information handling system and authenticate the client information handling system for use with the interim edge storage management system to access the edge storage device and the plurality of remote network file hosting systems managed by the interim edge storage management system; the processor executing computer readable program code of the interim edge storage management system that includes a storage orchestrator to manage selecting storage of the data files by the client device from among the edge storage device and the plurality of remote network file hosting systems via an edge local active directory of the edge storage device and plurality of remote network file hosting systems by selecting to store the data files accessed, updated, saved, or otherwise manipulated by the client device from among the edge storage device and the plurality of remote network file hosting systems based on recency of usage of the data files at the client device; and the processor executing computer readable program code of the storage orchestrator to gather data storage information related to the storage of one or more data files from a most recently used (MRU) data set to be stored on the edge storage device and one or more data files from a less recently used data set to be stored on the plurality of remote network file hosting systems; and the processor executing computer readable program code of the storage orchestrator to adjust the amount of the data stored on the edge storage device to at least one of the plurality of remote network file hosting systems based on one or more storage policies relating to storage capacity limits of the edge storage device by transitioning a portion of the data files from the edge storage device to the at least one remote network file hosting system for files that meet or exceed an access time-period threshold to transition those data files into the second less recently used dataset, wherein the first MRU data set includes data files accessed, updated, saved, or otherwise manipulated by the client information handling system that are different from the second less recently used data set of the client information handling system.

16. The storage management information handling system of claim 15 further comprising: the processor executing computer readable program code of the storage orchestrator to gather data storage information related to the storage of one or more data files of the second less recently used dataset stored on the plurality of remote network file hosting systems; and the processor to adjust the amount of the data stored on the plurality of remote network file hosting systems via transfer of the datafiles based on one or more storage policies relating to storage capacity limits of the remote network file hosting systems by transitioning data files from a first remote network file hosting system to a second remote network file hosting system based on storage capacity thresholds being reached at the first remote network file hosting system.

17. The storage management information handling system of claim 15 further comprising: the processor executing computer readable program code of the storage orchestrator to manage the storage and transitioning of the data files of the first MRU dataset onto the edge storage device based on the recency of usage of the data files at the client device as well as a storage policy of an application priority context of the data files from an application at the client device having an assigned priority to be stored at the edge storage device.

18. The storage management information handling system of claim 15 further comprising: the processor executing computer readable program code of the storage orchestrator to manage the storage of data files of the first MRU dataset by transitioning of one or more data files of the first MRU dataset onto the edge storage device based on detection of the recency of usage of the one or more data files at the client device and application of a storage policy relating to a frequency of access of the one or more data files.

19. The storage management information handling system of claim 15 further comprising: the processor executing computer readable program code of the storage orchestrator operatively coupled to the edge storage device storing and tracking data files of the first MRU data set

accessed, updated, saved, or otherwise manipulated by from the client device for storage on the edge storage device and enforcing rules for actively transitioning other data files falling into a second less recently used data set on the edge storage device to at least one remote network file hosting system based on one or more priority factors.

20. The storage management information handling system of claim 15 further comprising: the processor executing one or more application programming interfaces (APIs) executable for the storage orchestrator to interface with the plurality of remote network file hosting systems that are diverse remote network file hosting systems having file directories and formatting developed by a plurality of network file hosting services.
